

ABHISHEK DAS

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| Bangalore, India

SUMMARY

Software Engineer II with 6 years of experience building scalable, fault-tolerant distributed systems and event-driven microservices. Primary expertise in Java, Golang, and Kafka, with hands-on experience in PostgreSQL, Redis, and AWS. Drove backend projects end to end from system design to production across real-time IoT telemetry, enterprise compliance platforms, and high-throughput data pipelines. IIT Kanpur graduate (Minor in Computer Systems). Open to relocation.

EDUCATION

Indian Institute of Technology (IIT), Kanpur

Aug 2016 – May 2020

Bachelor of Technology | Major: Materials Science and Engineering | Minor: Computer Systems

PROFESSIONAL EXPERIENCE

PAYLOCITY (formerly Aidora)

Bangalore, India

Software Engineer II, Backend

Aug 2024 – Present

Eligibility and Tracking Team

- Built an **end-to-end versioned backend service** in **TypeScript and Next.js** managing leave datasets across all **50 US states** for **60+ enterprise clients**, ensuring accurate wage replacement and leave calculations across multiple regional jurisdictions.
- Engineered a **config-driven policy versioning framework** that auto-selects the correct policy version per employee based on leave start date or expected due date, requiring only a benefit config and an effective date with zero code changes, cutting onboarding time for annual US regulatory updates and reducing churn risk by **30%**, directly improving customer retention.
- Architected a **dynamic policy evaluation engine encoding 51 state-specific leave policies** across the 19 states and Washington DC that have their own paid leave laws, automating leave plan selection on top of federal baselines like FMLA and eliminating manual compliance effort for HR teams.
- Delivered backend support for **Minnesota Paid Leave** (effective 2026), handling dynamic bonding duration of 8 to 12 weeks based on disability, 52 or 53-week benefit years, and a shared family and medical leave pool with a 20-week annual cap. **Won a Minnesota-based enterprise client** on the strength of this implementation.
- Delivered backend support for **Delaware Paid Leave** (effective Jan 2026), handling asymmetric leave windows where parental leave allows 12 weeks per year while medical and family caregiving together allow 6 weeks per 24-month period, subject to a shared 12-week annual cap, and built employer size-based coverage tier logic to ensure correct eligibility per client from day one of the mandate.
- Designed **intermittent leave logic** covering medical, caregiver, and military scenarios, enabling non-continuous time off across all 60+ enterprise clients without breaking regulatory compliance for any jurisdiction.
- Built a **flexible reduced-schedule tracking module** handling partial-day increments and dynamic work schedules, eliminating manual HR calculations and ensuring precise benefit accrual across all supported jurisdictions.

ZOOMCAR

Bangalore, India

Software Engineer, Backend – Platform Team

Nov 2021 – Jul 2024

Real-time Communications and Notifications

- Built a **real-time two-way chat system** end-to-end using Java and Redis to resolve host-guest friction during vehicle handoffs, driving a **37% reduction in booking cancellations**.
- Integrated a scalable **text-to-speech voice alert system** via Java with AWS webhook tracking, achieving a **20% reduction in cancellation rates** across low-connectivity regions.
- Designed a **multi-channel communication orchestration system** escalating across email, push notification, and robocall when prior channels go unacknowledged within a configurable time window, reducing last-minute cancellations by **15%** and host-guest friction by **32%**.

Software Engineer, Backend – IoT Team

Nov 2021 – Jul 2024

Vehicle Telemetry and Real-time Data Infrastructure

- Architected and deployed a **high-throughput Golang and AWS-backed IoT gateway** as lead contributor, capturing and processing real-time vehicle telemetry from **12,000+ devices** across the Egyptian regional fleet.

- Engineered a **Bluetooth Low Energy (BLE)** integration between in-car IoT hardware and the mobile app using Java, enabling offline vehicle locking and unlocking and reducing **failed unlock attempts by 57%**.
- Designed an **event-driven Java and Kafka Streams microservice** to consume raw IoT messages and publish enriched, structured data to downstream analytics systems with PostgreSQL as the system of record, scaling throughput from **7 to 20 messages per second** as new device types were onboarded.
- Built a **real-time alerting pipeline** using Kafka state stores and Redis to proactively detect vehicle theft and security anomalies, notifying operations teams with sub-second latency and under **3% false positive rate**.
- Designed a **Kafka and PostgreSQL-backed data sync pipeline** to continuously replicate vehicle and device configuration changes across 20,000+ devices, supporting a **3x growth in fleet size** with no architectural changes.

INFERENCE LABS

Software Engineer

InfoX Team

Bangalore, India

Aug 2020 – Sep 2021

- Built a **scalable async job engine** in Golang leveraging MapReduce principles to execute heavy data processing jobs across large clinical datasets, cutting average job execution time for compute-intensive clinical workflows.
- Led the **architectural migration of the core wait-queue from Kafka to RocksDB**, eliminating throughput bottlenecks and significantly increasing capacity for large volumes of concurrent API requests.
- Implemented **Python Flask microservice APIs** across multiple services using the Protobuf message format for efficient, type-safe inter-service communication.
- Designed **automated Python benchmarking frameworks** to stress-test server performance across A100, V100, and T4 GPU clusters, enabling data-driven infrastructure decisions.
- Developed the full **React.js frontend** for a medical information extraction platform, enabling non-technical users to configure complex clinical text extraction rules without engineering support.

BOSCH GLOBAL

Software Engineering Intern

Cross-Platform Mobile Development

Bangalore, India

May 2019 – Jul 2019

- Built a **cross-platform mobile application** end-to-end using React Native and Firebase, integrating real-time location tracking via Google APIs and multi-provider authentication across Email, Facebook, GitHub, and Gmail.
- Developed core utility features including network diagnostics and real-time data display, delivering a production-ready application within the internship period.

TECHNICAL SKILLS

- **Languages:** Java, Golang, Python, C++, SQL, TypeScript
- **Backend and API:** REST, gRPC, Microservices, Event-Driven Architecture, Kafka Streams, Spring Boot
- **Frameworks:** Next.js, React.js, React Native, Flask
- **Messaging and Infrastructure:** Apache Kafka, Docker, Kubernetes, Linux
- **Databases and Caching:** PostgreSQL, MongoDB, RocksDB, Redis
- **Cloud and Observability:** AWS, Git, CI/CD

ACHIEVEMENTS

- **Exceeds Expectations, 3 consecutive half-yearly performance cycles** at Zoomcar, recognized for consistently high-impact engineering delivery across the Platform and IoT teams over a 2.5-year tenure.

PROJECTS

Ethereum-Based Student Senate Election Platform

Jan 2019 – Apr 2019

Deployed a Solidity smart contract on the Ethereum Ropsten Network to build a tamper-proof student election system, restricting voting access to verified IITK email holders using on-chain identity validation.

LANGUAGES

English: Full Professional Proficiency **Bengali:** Native